

电子游戏中可信度的挑战：定义、智能体模型与模仿学习

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摘要

本文探讨了在电子游戏中创建可信智能体（虚拟角色）的问题。我们仅考虑“可信度”的一种含义，即“给人一种由玩家控制的感受”，并概述了其评估问题。我们介绍了来自工业界和研究界的几种能够产生可信行为的游戏智能体模型。为了实现高水平的可信度，学习，尤其是模仿学习，似乎是可行之路。我们快速概述了让电子游戏智能体从玩家那里学习的方法。最后，我们提出了一种两步法来开发新的可信智能体模型。首先，我们必须为我们的应用找到可信度的标准，并定义一种评估方法。然后才能设计模型和学习算法。

关键词
自适应决策，可信度，类人，评估，模仿学习，电子游戏。

引言

如今，越来越多的游戏主机和电子游戏被设计出来，旨在让玩家感觉身临其境。为了定义这一目标实现的程度，学术界定义了两个标准：沉浸感和临场感。根据Slater的说法，沉浸感是一个客观标准，取决于硬件和软件（Slater等人，1995年）。它包括基于虚拟感官信息的类型、多样性、丰富性、方向性以及它们对真实感官信息的覆盖程度等标准。例如，力反馈和动作感应控制器、环绕声和高动态范围渲染可以提升沉浸感。临场感，也被称为远程临场感（Steuer 1992），是一个更主观的标准。它被定义为在环境中“身临其境”的心理感受，主要受电子游戏内容的影响。

正如(Slater等人，1995)所述，临场感部分依赖于感官数据与内部表征之间的匹配。

这种匹配体现了我们试图利用世界模型来更好地理解所感知到的事物，并能够进行预测(Held和Durlach, 1991)。这一理念与艺术领域中所说的可信度颇为接近。的确，我们之所以能够相信虚构的对象、地点、角色和故事，前提是它们大多符合我们的模型。提升电子游戏内容的可信度，进而应能增强临场感。

由于增强电子游戏可信度的方法众多，我们选择聚焦于可信虚拟角色，即可信智能体。做出这一选择的原因是，角色通常在书籍和电影故事的可信度中扮演着重要角色。然而，与书籍和电影角色不同，智能体应能应对各种可能的情况，而无需他人告知其该做什么。与其手动定义这些行为，让智能体自行学习这些行为会更有兴趣，这能缩短设计角色的时间。学习能力还具有提升可信度的优势，因此应被视为一项必备特性。

本文旨在概述在电子游戏中实现兼具可信度与学习能力的智能体所面临的各项可能性与约束。我们首先定义什么是游戏中的可信智能体，如何评估它们，以及可信度与电子游戏类型之间的关系。接着，我们简要概述智能体如何表达可信行为以及如何学习这些行为。最后，我们总结出一个实现可信智能体的两步协议。

可信度

可信度的定义

可信度这一概念极其复杂且主观。为了界定和理解这一概念，我们必须审视它在艺术中的含义——在艺术中，它是“搁置怀疑”的因素之一（Bates 1992）。据迪士尼两位核心动画师Thomas和Johnston所述，可信角色的目标是赋予观众“生命幻象”（Thomas和Johnston 1981）。而Reidl的定义则更为精确：

THE CHALLENGE OF BELIEVABILITY IN VIDEO GAMES: DEFINITIONS, AGENTS’ MODELS AND IMITATION LEARNING

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ABSTRACT

In this paper, we address the problem of creating believable agents (virtual characters) in video games. We consider only one meaning of believability, “giving the feeling of being controlled by a player”, and outline the problem of its evaluation. We present several models for agents in games which can produce believable behaviours, both from industry and research. For high level of believability, learning and especially imitation learning seems to be the way to go. We make a quick overview of different approaches to make video games’ agents learn from players. To conclude we propose a two-step method to develop new models for believable agents. First we must find the criteria for believability for our application and define an evaluation method. Then the model and the learning algorithm can be designed.

KEYWORDS

Adaptive decision making, believability, human-like, evaluation, imitation learning, video games.

INTRODUCTION

Nowadays, more and more consoles and video games are designed to make the player feel like he/she is in the game. To define how well this goal is achieved, two criteria have been defined in academic research: *immersion* and *presence*. According to Slater, immersion is an objective criterion which depends on the hardware and software (Slater et al. 1995). It includes criteria based on virtual sensory information’s types, variety, richness, direction and in which extend they override real ones. For example, force feedback and motion sensing controllers, surround sound and high dynamic range rendering can improve the immersion. Presence, also known as telepresence (Steuer 1992), is a more subjective criterion. It is defined as the psychological sense of “being there” in the environment. It is mainly influenced by the content of the video game.

As stated in (Slater et al. 1995), presence partly depends on the match between sensory data and inter-

nal representation. This match expresses the fact that we try to use world models to better understand what we perceive and to be able to anticipate (Held and Durlach 1991). This idea is close to what is called *believability* in the arts. Indeed, we can believe in fictional objects, places, characters and story only if they mostly fit in our models. Enhancing believability of video games’ content should then enhance the presence.

As there are many ways to enhance believability in video games, we choose to focus on believable virtual characters, also known as believable agents. The reason why we make this choice is because characters have often a major role in the believability of book and movie stories. However, unlike book and movie characters, agents should be able to cope with a wide range of possible situations without anyone to tell them what to do. Instead of defining manually these behavior, it can be interesting for the agents to learn them, reducing the time to design a character. The ability to learn has also the advantage of increasing believability so it should be considered as a must-be feature.

The goal of this paper is to have an overview of the possibilities and constraints for the realization of a believable and learning-capable agent in a video game. We first define what are believable agents in games, how we can evaluate them and the relations between believability and the type of video games. Then, we make a quick overview of how an agent can express believable behaviors and how it can learn them. Finally, we conclude on a two-step protocol to realize believable agents.

BELIEVABILITY

Definition of Believability

The notion of believability is highly complex and subjective. To define and understand this concept, we must look at its meaning in the arts where it is a factor of *suspension of disbelief* (Bates 1992). According to Thomas and Johnston, two core animators of Disney, believable characters’ goal is to provide the *illusion of life* (Thomas and Johnston 1981). Reidl’s definition is more precise:

“角色可信度指的是使角色能够实现‘生命幻象’的众多要素，包括但不限于个性、情感、意图性，以及生理学和生理运动”（Riedl和Young 2005，第2页）。Loyall则试图更客观地阐述，称这样的角色“提供了对其个性的令人信服的呈现，而这正是 /旁观者/所期望或逐渐期待看到的”（Loyall 1997，第1页）。这一定义与临场感的一个因素——即玩家的世界模型与感官数据之间的匹配——颇为接近。

如果我们试图将可信度的定义应用于电子游戏，情况会变得更加复杂。与古典艺术不同，玩家可以通过虚拟身体在游戏中具身化并进行交互。可信度的问题现在变成了：一个可信的角色是必须赋予生命幻象，还是必须让玩家觉得它是由另一位玩家操控的？（Livingstone 2006）。即使电子游戏描绘的是现实世界，这两者之间也可能存在巨大差异，因为一切都是虚拟的，玩家知道自己的行为没有实际后果。

在本文中，我们仅将“可信”定义为“给人以被玩家操控的幻象”。乍看之下，这似乎与临场感相悖，因为它提醒了玩家现实世界的存在。然而，不采用这种定义也存在一些弊端：仅赋予生命幻象的虚拟角色可能会被归类为“一段程序”，这反而可能永久性地打破幻象。玩家也可能仅仅因为知道虚拟角色并非由人类操控，而察觉到其行为中的问题。

既然我们已经选择了可信度的定义，就需要找出影响它的不同标准。明确这些标准将使改进和评估成为可能。

可信度标准

定义可信度的标准高度依赖领域。例如，对于具身对话代理（ECA），语音质量、面部表情、手势等因素都很重要（Ruttkay 等人 2002）。对于情感和社会代理，对他人的反应是最关键的部分（Reilly 1996），而研究可信智能体的众多领域亦是如此。然而，存在一个适用于所有领域的标准。所谓的 ELIZA 效应（Weizenbaum 1966）非常有趣：只要智能体没有主动打破幻觉，人们往往会从中看到并不存在的复杂思维机制。

摧毁这种错觉的一个方法是可预测性，这是电子游戏中角色常见的缺陷。

那些一遍又一遍重复完全相同行为的角色，很快就会被归类为虚假的。增加一些不可预测性可以大大提升可信度（Bryant和Miikkulainen 2006）。难点在于，过多的不可预测性又可能带来随机感，同样会损害可信度。另一个缺陷是明显的学习失败。如果智能体在行为明显适得其反时仍保持不变，就会打破其“智能”的假象。这正是我们认为可信智能体必须能够学习的主要原因（Gorman和Humphrys 2007）。

我们将引用的最后一个标准区分了可信度与现实主义。与现实主义智能体不同，可信智能体可能不得不刻意夸张，以便观察者理解其行为（Pinchbeck 2008）。尽管这看起来有些奇怪，但有时必须夸大智能体的某些特征，才能让人们相信它。这种技巧在艺术中非常常见，尤其在卡通作品中。这意味着类人智能体的可信度可能相当低。然而，现实主义与可信度之间存在联系，因此从现实主义中汲取灵感不失为一个良好的起点。

了解哪些因素影响可信度后，我们就可以设计针对可信智能体的评估方法。由于上述标准较为笼统，评估时还需考虑更多领域特定的标准。

可信度评估

由于可信度具有主观性，评估是一个关键且复杂的步骤。尽管图灵测试（Turing 1950）最初并非为此目的设计，但它仍被视为可信度评估的参考基准。在标准解释中，评判员需要仅通过文本与人类和机器进行对话。如果经过一定时间后，评判员无法分辨哪个是人工智能，则该机器被认为具有智能。这一测试旨在评估智能，但备受批评（Searle 1980; Hayes and Ford 1995）。然而，这些批评并不适用于可信度评估，相反，它甚至是我们此前定义的可信度评估的一个极佳基础。

可信度评估方法涉及诸多参数（Mac Namee 2004; Gorman 等 2006; Livingstone 2006）。首要参数是是否要对所呈现角色的本质提出质疑。这一选择通常将与智能体和人类混合有关，以便评判者也能评估真实人类的可信度。这样做有助于避免由偏见引起的偏差，并提供一个参考：人类通常不会获得完美的可信度评分。另一个参数是问答数量。图灵测试仅包含一个

“*Character believability refers to the numerous elements that allow a character to achieve the ‘illusion of life’, including but not limited to personality, emotion, intentionality, and physiology and physiological movement*”（Riedl and Young 2005, page 2）。Loyall tries to be more objective saying that such a character “*provides a convincing portrayal of the personality they [the spectators] expect or come to expect*”（Loyall 1997, page 1）。This definition is quite close to one factor of the presence, the match between players’ world model and sensory data.

If we want to apply the believability definition for video games, things become even more complex. Unlike classic arts, players can be embodied in a game by the mean of virtual bodies and can interact. The believability question is now: does a believable character have to give the illusion of life or have to give the illusion that they are controlled by a player?（Livingstone 2006）。There can be very important differences as even if the video game depicts the real world, all is virtual and players know that their acts have no real consequence.

In this paper, we will consider only believable as “giving the illusion of being controlled by a player”. At first glance, it can be seen as going against presence as we remind the players that there is a real world. However, not using this approach has some drawbacks too: virtual characters only giving the illusion of life can be classified as “being a piece of program” which may break the illusion permanently. Players may also see problems in the virtual characters’ behavior only because they know they are not human-controlled.

Now that we chose a definition for believability, we have to find the different criteria that influence it. Defining those criteria will make improvements and evaluation possible.

Believability Criteria

Criteria to define believability are highly domain-dependant. For example, for embodied conversational agents (ECA), factors like quality of the speech, facial expressions, gestures are important (Ruttkay et al. 2002). For emotional and social agents, reaction to others is the most important part (Reilly 1996) and so on with the numerous domain working on believable agents. However, there is one criteria that can be used for every domains. What have been named Eliza effect (Weizenbaum 1966) is really interesting: as long as the agent is not actively destroying the illusion, people tend to see complex thinking mechanisms where there are not.

One way to destroy the illusion is predictability, which is a classic flaw for characters in video games. Char-

acters doing over and over the exact same thing are rapidly categorized as being artificial. Adding some unpredictability can better a lot believability (Bryant and Miikkulainen 2006). The difficulty is that too much unpredictability can give the feeling of randomness which can harm believability too. Another flaw is obvious failure to learn. An agent keeping the same behavior even if it is clearly counter-productive breaks the illusion of being “intelligent”. That is the main reason why we think that believable agents must be able to learn (Gorman and Humphrys 2007).

The last criterion we will cite makes the difference between believability and realism. Unlike realistic agents, believable agents might be forced to overdo for observers to understand what they are doing (Pinchbeck 2008). Although it can seem strange, it is sometimes mandatory to exaggerate some of the agent’s characteristics so that people believe in it. This technique is very often used in arts, especially in cartoons. This means that human-like agents could have a quite low believability. There are however links between realism and believability so it should be a good start to draw inspiration from realism.

Knowing what influence believability, we can design an evaluation method for believable agents. As the cited criteria are quite general, more domain-specific ones should be taken into account in the evaluation.

Evaluation of Believability

As believability is subjective, evaluation is a critical and complex step. Even if it was not intended to, Turing’s test (Turing 1950) is still considered as a reference for believability evaluation. In its standard interpretation, a judge must chat with a human and a machine using only text. If, after a certain amount of time, the judge cannot tell which one is artificial, the machine is said to be intelligent. This test’s goal was to assess intelligence but it has been much criticized (Searle 1980; Hayes and Ford 1995). This critique, however, does not apply to believability and it is even a very good basis for assessing believability as we defined earlier.

There are many parameters for believability evaluation methods (Mac Namee 2004; Gorman et al. 2006; Livingstone 2006). The first one is to cast or not to cast doubt on the nature of the presented character(s). This choice is often linked with mixing agents and humans so that the judges assess real humans’ believability too. This can be useful to avoid bias induced by prejudices and to have a reference: humans usually do not score a perfect believability. Another parameter is the number of questions and answers. Turing’s test features only one

问题和是/否回答，而其他测试则包含多个问题和评分量表。前者可能过于局限，而后者则可能导致太多未确定的答案来“通过”测试。另一个问题在于整体可信度评分的计算，在有多道题目的情况下，可能会给实验者带来过大的结果影响。为了增加客观性，可以采用相对评分而非绝对评分：给某个示例的评分可以回答“示例A是否比示例B更好？”这样的问题。此外，还需要决定评判者是玩家还是旁观者。玩家可以主动测试被评估的角色，而旁观者则更专注于这些角色，能够注意到行为中更多的细节。最后，评判者的选择至关重要。文化背景（Mac Namee 2004）和经验水平（Bossard 等人 2009）可能会对可信度评分产生显著影响。

可信度与电子游戏之间的关联

根据我们对可信度的定义，多名玩家应能在同一款电子游戏中共同游玩。实际上，玩家不得事先知道他们所感知到的是由其他玩家还是由程序产生的。可信智能体应被用来替代缺失的玩家，并扮演与其他玩家相同的角色，这样我们才能实现高水平的可信度。当然，单人游戏可以打造类人智能体，但按照我们的定义，它们并不具备可信性。

电子游戏中交互的复杂性对可信度有着非常重要的影响。如果玩家、智能体与游戏之间的交互很少，那么评估可信度可能会很困难。例如在乒乓球游戏中，玩家只能移动球拍，因此他们可能很难猜测对手的想法。相反，如果玩家和智能体拥有广泛的可选行动，他们就能更好地感知对方的意图，但这反而使得可信度更难实现。因此，应优先选择那些具有复杂世界并允许大量交互的电子游戏来开发和评估可信智能体。

电子游戏通常被分为不同的类别。作为测试平台，我们需要一款支持玩家间大量互动的多人游戏。这类游戏可以在动作、角色扮演、冒险和体育游戏类别中找到。从技术角度来看，冒险游戏和体育游戏往往难以修改，尤其难以添加智能体。角色扮演类游戏涉及大量沟通，而自然语言过于复杂，难以让对话智能体真正令人信服。动作游戏通常是不错的选择，因为它们的动作集相当有限，简化了学习过程，但其组合方式却足以构成可信度方面的挑战性难题。

其复杂度足以成为可信度方面的挑战性问题。

此类动作游戏的经典例子是第一人称射击游戏。在这些游戏中，每位玩家或智能体控制一个独特的虚拟身体，并通过其眼睛进行观察。角色可以（非详尽地）拾取物品（武器等）、移动（行走、跳跃等）以及用武器射击。每个角色都有一定数量的生命值，也称为生命点数：每当一个行动者被敌方火力击中时，当前生命值就会减去一定数量。当生命值归零时，角色会“死亡”，但通常可以在游戏环境中的另一个位置重新出现。尽管这个概念看似非常简单，但要设计出可信智能体却颇具挑战性。为了在开始时简化问题，可以避免使用合作规则，因为所采用的战术可能相当难以学习。

智能体行为建模

智能体的通用原理是感知其周围环境，根据这些感知及其内部状态做出一些决策，然后根据决策执行一些行动。这被称为感知-决策-行动循环，因为该序列会不断重复。行动和感知的集合由电子游戏定义，因此设计一个智能体“只是”找到感知与行动之间的良好映射关系。

智能体思考机制的一般结构称为模型。模型在参数化后生成行为：当输入感知序列时，它会输出动作序列。换句话说，模型是表达行为的方式，而参数则是行为本身。

存在许多不同的智能体行为模型，其中一些模型能够比其他模型实现更高的可信度。有三个领域对其设计影响深远：计算机科学、心理学和神经科学。计算机科学将智能体模型视为程序，具有输入和输出，并由数学函数、循环等组成。受心理学影响的模型则试图应用科学对人类行为的理解。最后，基于神经科学的模型试图构建一个由不同专门区域中的神经元组成的人工大脑。存在许多混合模型，甚至有些模型将上述三者全部融合。

在智能体的思考方式上，也存在两种不同的路径：反应式或慎思式。反应式智能体直接将行动与感知相映射。问题在于，它们往往缺乏长期目标的行动能力。而慎思式智能体则试图预测

question and a yes/no answer whereas other tests feature many questions and scales to answer. The former choice may be too restrictive while the latter may result in too much undecided answer to “beat” the test. Another problem is the computation of the overall believability score which, in case of multiple questions, may give experimenters too much influence on the results. To add more objectivity, it is possible to have relative scoring instead of absolute: the score given to a example can answer to “is example A better than example B?”. It is also necessary to decide if judges are players or only spectators. While players can actively test evaluated characters, spectators are more focused on them and can notice much more details in the behaviors. Finally the choice of the judges is really important. Cultural origins (Mac Namee 2004) and level of experience (Bossard et al. 2009) may have a noticeable impact on believability scores.

Links between Believability and Video Games

Following our definition of believability, multiple players should be able to play together in the same video game. Indeed, players must not know *a priori* if what they perceive is due to another player or to a program. Believable agents should be used to replace a missing player, having the same role as the others so that we can achieve high level of believability. Of course, single player games can achieve human-like agents but they will not be believable as we defined it.

The video games’ complexity of interactions has a really important impact on the believability. If interactions between the players, agents and the game are few, it may be hard to assess believability. For example in a pong game, players can only move paddles so it may be difficult for them to guess the thoughts of the adversary. On the contrary, if players and agents have a wide range of possible actions, they can better perceive the others’ thoughts making believability harder to achieve. Video games featuring a complex world and allowing a lot of interactions should be preferred to develop and evaluate believable agents.

Video games are often categorized in different groups. For a testbed, we need a multiplayer game with a lot of possible interactions between players. This kind of games can be found in the action, role playing, adventure and sport games categories. From a technical point of view, adventure and sport games tend to be difficult to modify and in particular to add agents. Role playing involves a large part of communication and natural language is far too much complex to make conversational agents truly believable. Action games are often good choice because they have a quite limited set of actions, simplifying the learning, but their combination are com-

plex enough to be a challenging problem for believability.

The classical example of such action games is first person shooter games. In those games, each player or agent controls a unique virtual body and sees through its eyes. The character can, non-thoroughly, grab items (weapons, ...), move (walk, jump, ...) and shoot with a weapon. Each character has an amount of hit points, also known as life points: each time an actor is hit by an enemy fire, a certain amount of hit points are subtracted to the current value. When hit points reach zero, the character “dies” but can usually reappear at another place in the game’s environment. Although the concept can seem very basic, it can prove challenging to design believable agents. To make things simpler at the beginning, it is possible to avoid cooperative rules as tactics used may be quite hard to learn.

AGENTS’ BEHAVIOUR MODELLING

The general principle of an agent is to perceive its surroundings, take some decisions according to those perceptions and its internal state and then make some actions which depend on the decisions. This is named the perception, decision, action loop because the sequence is repeated over and over. The sets of actions and perceptions are defined by the video game so designing an agent is “only” to find a good mapping between perceptions and actions.

The general structure of the agents’ thinking mechanisms is called a model. The model, when parametrized, generates behaviors: it gives a sequence of actions when fed with a sequence of perceptions. In other words, the model is the way to express behavior and the parameters are the behavior.

There are many different agents’ behavior models and some can achieve better believability than others. Three domains are very influential on their design: computer sciences, psychology and neurosciences. Computer sciences see agents’ models as programs, with inputs and outputs and composed of mathematical functions, loops, etc. Models influenced by psychology try to apply what science understood about humans’ behaviors. Finally, models based on neurosciences try to code an artificial brain composed of neurons grouped in different specialized zones. There are many hybrid models and even some of them combines all the three.

There are also two different approaches in the way the agents think: reactive or deliberative. Reactive agents map directly actions to perceptions. The problem is that they tend not to act with long term objectives. Deliberative agents, on the other hand, try to predict

其行动的结果。它们制定计划，使一系列行动能够满足自身需求。其主要弱点在于，在快速变化的环境中往往表现不佳。部分智能体将反应式与慎思式能力相结合，从而兼取两者优势，规避各自弱点。

工业界中的模型

在电子游戏工业界中，模型通常相当基础，但在熟练的游戏设计师手中却能够实现良好的可信度。有限状态机、层次有限状态机和行为树被广泛使用。这些模型能够表达逻辑、执行基本规划，并且擅长处理反应式行为。另一个优势在于这些模型是确定性的，换句话说，我们可以预测可能发生的事情，从而尽量避免问题。它们的主要弱点是表达能力有限，对于复杂行为的解释可能变得难以管理。

在电子游戏工业界中使用的另一种模型是规划器。它们在可信度方面能带来非常不错的结果，一个例子是游戏 F.E.A.R.，它部分基于 STRIPS 规划器（Fikes 和 Nilsson, 1971）。配备规划器的智能体能够制定复杂的战术，这对可信度有重要影响。然而，它们尚不具备学习能力，因此可能变得可预测。其他模型在电子游戏中使用得较少。

研究中的模型

在科学研究中，智能体使用了多种不同的模型。一些主要的架构包括 Soar (Newell 等人 1987)、ACT-R (Anderson 1993) 和 BDI (Rao and Georgeff 1995)。然而，这些模型并非为可信度而设计，而是用于复现人类的某些思维机制。有些模型从一开始就为可信度而设计，例如 (Lester and Stone 1997; Mac Namee 2004) 以及 Oz 项目 (Bates 1992; Reilly 1996; Loyall 1997) 中使用的模型，但这些模型并不适用于动作游戏，而是适用于情感和社交互动。

在动作游戏领域，已有一些相关研究并提出了不同的模型。(Le Hy 等人, 2004) 提出的贝叶斯模型似乎能生成相当可信的行为，但缺乏长期目标。(Robert 和 Guillot 2005) 提出的模型使用带有在线无监督学习算法的分类器，但智能体的可信度尚未得到评估。(Gorman 等 2006) 部分基于 (Rao 等人 2004) 的工作，采用贝叶斯模仿，实现了良好的可信度。然而，它仅生成了游戏所需完整行为的一部分。(Gorman 和 Humphrys 2007) 模型“仅”基于神经

网络，但由于其学习算法，它在可信度方面取得了非常好的效果。同样，它也只生成了行为的一部分。

采用混合反应式-慎思式智能体似乎是一个很好的选择，因为规划对可信度影响巨大，反应性同样如此 (Laird and Duchi 2000)。与电子游戏中常用的做法相反，非确定性方法更可取，因为它可以通过避免重复性来提高可信度。智能体必须能够学习，并且在最佳情况下，应只需最少的先验知识。这意味着该模型无需易于理解和调整，因为其大部分知识可能来自学习。

动作与感知

模型输入与输出的选择至关重要，因为它对最终结果有着巨大影响。合乎逻辑的做法是让智能体与玩家拥有完全相同的感知与动作信息集合，这样智能体才能拥有成为可信角色所需的一切工具。然而，感知与动作信息在玩家大脑与游戏之间遵循着一个复杂的过程。此外，采用类人交互方式可能会使问题更加复杂，非但无法提升可信度，反而可能适得其反。

感知信息始于电子游戏内部，游戏中的每个对象都拥有一组特征，如位置、旋转、颜色或纹理。所有对象经过渲染后，形成像素序列。这些像素通过屏幕或视频投影仪等外设显示出来。接着，信息传入我们的眼睛，再到达我们的大脑。我们以视觉信息为例，这一过程同样适用于声音、触觉，未来也将适用于气味和味觉。

正如我们所见，感知信息形式多样，但并非所有信息都对具有可信行为的智能体适用或有用。从游戏环境被渲染的那一刻起，我们就进入了人工视觉、听觉等领域。遗憾的是，这是一个非常棘手的问题，通过这种方式能获得的信息极少。因此，在渲染完成时，可用信息会大量丢失。最佳的折衷方案是直接使用来自游戏的信息。这样既能获得非常精确的信息（如位置），也会遇到无法获取的信息（如光照）。

对于动作而言，它们起始于我们的大脑，随后经过肌肉，通过游戏的外设，最终抵达游戏世界。虽然可以模拟外设的活动，但这也有其弊端。外设信息属于非常低级的序列，因此我们必须将高级

the results of their actions. They make plans so that the sequence of their actions will satisfy their needs. Their main weakness is that they are often not very good in rapidly changing environments. Some agents combine reactive and deliberative abilities to have both advantages without the weaknesses.

Models in Industry

In the video game industry, models are often quite basic but can achieve good believability in the hands of skilled game designers. Finite state machines (FSM), hierarchical FSM and behaviors trees are widely used. Those models can express logic, basic planning and are good at reactive behaviours. Another advantage is that those models are deterministic, in other words we can predict what can happen and then try to avoid problems. Their major weakness is that their expressiveness is limited and explaining complex behaviors can become unmanageable.

Another type of model which is used in video game industry is planners. They can give very good results in terms of believability, an example is the game F.E.A.R. which is partially based on the STRIPS planner (Fikes and Nilsson 1971). Agents featuring planners are able to set up complex tactics which have an important impact on believability. However they are not yet capable of learning so they can become predictable. Other models are, by less, used in video games.

Models in Research

In scientific research, many different models are used for agents. Some of the main architectures are Soar (Newell et al. 1987), ACT-R (Anderson 1993) and BDI (Rao and Georgeff 1995). However, these models are not design for believability but for reproducing some humans’ thinking mechanisms. Some models are, from the beginning, designed for believability like those used in (Lester and Stone 1997; Mac Namee 2004) and in the Oz project (Bates 1992; Reilly 1996; Loyall 1997) but they are not suited for action games but for emotions and social interactions.

For action games, some work has been done and different models have been presented. (Le Hy et al. 2004) is a Bayesian model which seems to produce quite believable behaviors but lacks of long-term goals. (Robert and Guillot 2005) is a model using classifiers with a online unsupervised learning algorithm but the agents’ believability has not been evaluated. (Gorman et al. 2006) use Bayesian imitation, partly based on (Rao et al. 2004), and have a good believability. However it only produces a part of the whole behavior needed for the game. (Gorman and Humphrys 2007) is “only” based on neu-

ral networks but due to its learning algorithm, it gives really good results in terms of believability. Here too, it produces only a part of the behavior.

A good choice seems to have an hybrid reactive-deliberative agent as planning has a great impact on believability so as reactivity (Laird and Duchi 2000). Contrary to what is commonly used in video games, a non-deterministic approach is preferable because it can increase believability by avoiding repetitiveness. Agents must be able to learn and, in the best case scenario, should need minimal *a priori* knowledge. This implies that the model does not need to be easily understandable and adjustable as most of its knowledge could come from learning.

Actions and Perceptions

The choice of the model’s inputs and outputs is very important because it has a great influence on the final result. What seems logical is to have the exact same sets for the agents and for the players so that agents have all the necessary tools to be believable. However, the perceptions and actions information follow a complicated process between the players’ brain and the game. Moreover, having human-like interaction can make the problem even more complicated and instead of improving believability, it could make it worse.

The perception information begins in the video game, each object in it having a set of characteristics like position, rotation, color or texture. All those objects are rendered so they are a sequence of pixel. Those pixels are displayed by a peripheral like a screen or a video projector. Then the information go to our eye and then to our brain. We use visual information as an example, it is also true with sounds, touch and will be true for odors and tastes in the future.

As we can see, the perception information takes multiple forms and all may not be usable or useful for believable agents. From the moment the game’s environment is rendered, we enter the domain of artificial vision, audition, etc. Unfortunately, this is a very difficult problem and very few information can be obtained this way. As a consequence, there is a great loss of available information when the rendering is done. The best compromise is to use information directly from the game. It results in having both very precise information, like position, and unavailable information, like lighting.

For the actions, they begin in our brain, then muscles, go through the game’s peripherals and arrive in the game. It is possible to simulate peripherals’ activity, however it has some drawbacks too. Peripheral information are very low-level so we must translate high-level

目标转化为低级序列。如果智能体想要前往某个地点，它必须找到一系列类似外设的动作序列，而不是为其虚拟身体定义一条运动轨迹。

通过模仿学习

为何要模仿？

从一开始，我们只讨论了学习，而未具体说明“如何”学习。我们之所以选择专注于模仿，是因为**可信度**。正如我们所定义的，可信度意味着“看起来像玩家”。实现这一目标的有效方法之一，便是复制或模仿他/她 (Schaal 1999)。

定义与方法

模仿在“要创新，不要模仿”这类说法中带有相当负面的含义。然而，它是一种巧妙的学习方式，只有极少数动物能够使用 (Blackmore 1999)。此外，人类经常使用模仿，频繁到常常自己都未察觉，并且从生命之初就开始这样做 (Meltzoff and Moore 1977)。模仿是观察某种**行为**并大致重复该行为的过程。这种重复可以延迟进行，并且**示范者**可能知晓也可能不知晓自己被模仿。

从行为与认知科学的角度来看，模仿是执行先前被示范行为的倾向性增强。被模仿的行为必须对模仿者来说是新的，且任务目标应与示范者相同 (Schaal 1999)。关于策略是否应当相同，存在一些争议。根据 Meltzoff 的观点，推断意图对于高级模仿是必要的，在高级模仿中，模仿者学习的是目标而非实现目标的策略 (Rao 等人 2004)。这可以使得即使是不成功的尝试，只要目标能被推断出来，也能作为模仿学习的示例。在这种情况下，策略可能与示范者给出的示例有所不同。

从神经科学的角度来看，可以通过研究进化动物的大脑来解释模仿行为。根据对猕猴大脑的研究，某些区域被认为参与模仿学习过程。近年来镜像神经元的发现可能有助于理解模仿的机制。镜像神经元是指当我们自己做某事或看到别人做同样事情时都会放电的神经元。然而，目前尚不能确定它们在模仿过程中是否真正扮演主要角色。

在机器人学和人工智能领域，模仿学习正变得越来越流行。它尤其在机器人运动模仿中得到应用，因为这减少了类人机器人的编程时间 (Gaussier 等人 1998)。确实，当我们向他人解释一个动作时，通过示范远胜于仅用语言解释。对于机器人和智能体也是如此：模仿缩小了程序需要搜索的假设空间。

计算机科学中的模仿学习

我们选择模仿学习的主要原因在于，可信度与模仿对象（在我们的案例中为人类）的相似性密切相关。一些学者认为，模仿可以产生可信的、至少是类人的智能体和机器人 (Schaal 1999; Thureau 等人 2005)。这种学习技术似乎既快速又适合我们的目标。

Rao和Meltzoff (Rao 等人 2004) 设计了一种基于模仿的有趣模型和学习算法。该模型之所以特别引人注目，是因为它基于Meltzoff关于婴儿模仿的研究。该模型使用概率来确定应根据感知采取哪些行动。这些概率是通过观察玩家的虚拟身体习得的。另一个有趣的特点是，智能体可以猜测玩家的目标，以更好地理解其行为。该模型和算法已得到扩展，并应用于一款动作游戏 (Gorman 等 2006)，取得了有趣的结果。

一些研究更侧重于游戏中动作的模仿，这些动作通常同时具有高度反应性和策略性。在他们的工作中，Thureau等人使用一种名为神经气的算法来表示游戏环境的几何结构 (Thureau等人 2004)。然后，他们利用一种算法，通过给出玩家所走的路径来学习每个神经元/航点的潜力。这些潜力的总和形成一个力场，将智能体吸引到有趣的地方。

另一项有趣的研究并未采用新模型或新算法 (Gorman和Humphrys 2007)，且仅涵盖一个行为方面：动作游戏中的瞄准与射击。该模型由三个神经网络组成，一个用于选择武器，一个用于瞄准，另一个用于决定是否开火。这些神经网络使用Levenberg-Marquardt算法，基于先前收集和处理过的玩家数据进行训练。结果非常令人鼓舞，智能体能够进行短期预判，并以真正类人的方式行动。智能体甚至模仿了因使用鼠标而产生的行为：惯用右手的玩家在

goals to low-level sequences. If the agent wants to go to some place, it has to find a sequence of peripheral-like actions instead of defining a trajectory for its virtual body.

LEARNING BY IMITATION

Why Imitation?

Form the beginning, we have discussed only about learning without specifying the “how”. The reason we choose to focus on imitation only is because of believability. As we defined it, believability means “to look like a player”. One efficient way to achieve this is to copy or imitate him/her (Schaal 1999).

Definition and Approaches

Imitation has quite a pejorative connotation as the “innovate, don’t imitate” sentence shows. It is however a clever way of learning and very few animals can use it (Blackmore 1999). Moreover, humans use it very often, such often that they do not always notice it and they do it from the very beginning of their lives (Meltzoff and Moore 1977). Imitation is the act of observing a certain behavior and roughly repeating it. This repetition can be deferred and the demonstrator may or may not be aware of being imitated.

From a behavioral and cognitive sciences approach, imitation is the increased tendency to execute a previously demonstrated behavior. The imitated behavior must be new to the imitator and the task goal should be the same as the demonstrator (Schaal 1999). There are some debates on whether or not the strategy should be the same. According to Meltzoff, inferring intentions is necessary for higher level imitation where imitators learn the goals instead of the strategy to achieve them (Rao et al. 2004). This can lead to unsuccessful attempts being use as examples in imitation learning as long as the goal can be inferred. It this case, the strategy may be different from the example given by the demonstrator.

From a neuroscience approach, imitation can be explained studying the brains of evolved animals. Some areas are hypothesized to be involved in imitation learning according to studies on macaques’ brains. The quite recent discovery of *mirror neurons* may help in understanding the mechanisms of imitation. Mirror neurons are neurons that fire both when we do something and when we look at someone doing the same thing. However it is not sure that they actually have a major role in the imitation process.

In the domains of robotics and artificial intelligence, imitation learning becomes more and more popular. It has been particularly used in movements imitation in robotics because it reduce the programming time of humanoid robots (Gaussier et al. 1998). Indeed, when we explain a movement to somebody else, showing the example is far more productive than explaining only with words. The same goes with robots and agents: imitation reduce the space of hypothesis the program has to search.

Imitation Learning in Computer Sciences

The main reason we chose imitation learning is that believability is quite linked with resembling to the demonstrator, in our case humans. According to some authors, imitation can lead to believable or at least humanoid agents and robots (Schaal 1999; Thureau et al. 2005). This learning technique seems to be both fast and adequate for our goal.

An interesting model and learning algorithm based on imitation has been designed by Rao and Meltzoff (Rao et al. 2004). What makes it particularly interesting is that it is based on Meltzoff’s work on imitation in infants. The model uses probabilities to determine which actions should be taken according to the perceptions. Those probabilities are learnt observing a player’s virtual body. Another interesting characteristic is that the agent can guess the player’s goals to better understand his/her behavior. This model and algorithm have been extended and used in a action game (Gorman et al. 2006) leading to interesting results.

Some work focused more on the imitation of movement in games which is often highly reactive and strategic at the same time. In their work, Thureau *et al.* used an algorithm named neural gas to represent the geometry of the game’s environment (Thureau et al. 2004). They then use an algorithm to learn potential for each neuron/waypoint by giving path followed by players. The sum of those potentials form a field force, attracting the agent to interesting places.

Another interesting work does not use a new model or a new algorithm (Gorman and Humphrys 2007) and it only covers one behavioral aspect: aiming and shooting in an action game. The model consists of three neural networks, one for choosing a weapon, one for aiming and one to choose to fire or not. Those neural networks are trained with a Levenberg-Marquardt algorithm on players’ data previously gathered and treated. The results are very encouraging as the agents were capable of short-term anticipation and acted in a really human-like fashion. Agents even copied behaviors due to the use of a mouse: right-handed players have more difficulties to

用光标追踪从右向左移动的目标时，会面临更多困难。

最后一种基于模仿学习且效果显著的方法是 (Le Hy 等人, 2004)。对于基本运动控制，它使用继承规则通过统计出现次数来找出动作概率。该规则的主要价值在于能够估计未出现动作的概率。对于需要完成的任务连接，由于模型接近隐马尔可夫模型（HMM），因此采用增量式Baum-Welch算法来求解参数。该算法支持在线学习，因此可能成为提升可信度的有力工具。其主要缺点在于必须找到一种方法来识别当前任务，以便修改相应的参数。

学习约束

如前所述，明显的学习失败会破坏生命幻象。我们的第一个约束是学习过程必须是可观察的。这意味着智能体能够在游戏过程中进行学习，并且学习速度要快。第一个要求在人工智能领域被称为在线学习，即模型针对每个给定的示例进行修改。这与离线学习形成对比，后者仅在所有示例都给出后才修改模型。

尽管模仿学习可能是一种快速的学习方式，但这一目标并不容易实现。模仿学习的难点在于泛化：我们不希望智能体只是示范者所展示内容的“底片”，而是能像示范者那样适应环境。学习系统通常需要正例和反例来学习，以避免过度泛化和过学习。在模仿学习中，我们只有正例，因此可能需要大量样本，才能让智能体理解如何进行泛化。这可能会大大减缓学习速度。

如前所述，学习算法与智能体的行为模型紧密相连。实际上，学习的过程就是修改模型参数的过程。算法甚至可能改变模型的结构。选择或设计学习算法时，不能脱离行为模型的选择或认知。这使得选择更加复杂，因为我们必须考虑这对组合的利弊，而非单独评估每一个。

结论

在本文中，我们概述了在电子游戏中创建可信智能体的问题。该问题涉及三个参数：玩家、电子游戏和智能体。

我们可以在此基础上加入学习机制，使智能体具备演化能力。所有这些因素都会对最终结果产生重要影响，因此应仔细研究它们的作用。

首先，可信度因其主观性而成为一个非常复杂的概念。我们将其定义为“智能体由玩家控制的幻觉”。尽管这一定义与艺术领域的定义截然不同，但在电子游戏中却具有意义。事实上，多人可以在同一款游戏中互动，而用计算机程序替代其中部分玩家可能非常有趣。使可信度复杂化的原因在于，尽管玩家难以理解他人的行为且经常错误解读，但一些细微的缺陷便足以摧毁整个幻觉。

另一个重要因素是电子游戏本身。在仅能聊天的游戏中，可信智能体与在仅能移动的游戏截然不同。那些要求同时具备反应式与规划能力，并提供玩家之间广泛互动可能性的复杂游戏，对研究界而言最具挑战性的测试平台。

智能体本身，更确切地说是定义其行为的模型，是该问题中最显著的因素。模型能够表达不同程度可信行为，取决于其对人类特征的建模方式。因此，尽管研究界和工业界都存在诸多模型，但并非所有模型都适用于构建可信智能体。

如果一个智能体不具备演化的能力，它将无法长时间维持这种幻象。演化的一个可能途径是学习。模仿学习似乎与可信度这一目标完美契合。实际上，它通过复制示范者行为来实现，这与可信度的概念非常接近。同时，这也是人与人之间进行知识传递的一种非常快速的方法。

设计可信智能体的第一步是定义一种评估方法。这将帮助我们了解哪些特征对我们的智能体至关重要。但这一工作无法在选定一款电子游戏之前完成，因为可信度的概念取决于智能体的行为集。此外，先对人类和简单智能体的可信度进行评估，以更好地理解该问题，也是极具价值的研究方向。

接下来的步骤是选择或设计一个行为模型和一个模仿学习算法。这应根据上一步中发现的可信度因素来进行。考虑到快速学习的约束，

follow targets travelling from the right to the left with their cursor.

One last work based on imitation learning which seems to give very good results is (Le Hy et al. 2004). For basic motion control it use the rule of succession to find probabilities of action by counting the number of occurrences. This rule’s main interest is that it estimates the probabilities of actions that did not occur. For the linking of the tasks to achieve, as the model is close to a hidden markov model (HMM), an incremental Baum-Welch algorithm is used to find the parameters. This algorithm permits online learning so it may prove to be a great tool for believability. The main drawback is that we must find a way to identify the current task so that corresponding parameters can be modified.

Learning Constrains

As we said earlier, obvious failure to learn can break the illusion of life. Our first constraint is that the learning process can be observable. This implies that the agent can learn while playing and that it learns fast. The first implication is named *online learning* in terms of artificial intelligence. It means that the model is modified for each given example. It contrasts with *offline learning* where the model is only modified when all examples are given.

Despite the fact that imitation learning can be a fast way to learn, this goal is not easy to achieve. The difficulty in imitation learning is generalization: we do not want the agent to be a film of what the demonstrator showed but to adapt to the environment as the demonstrator did. Learning systems often need examples and counter-examples to learn to avoid over-generalization and over-learning. In imitation learning we only have examples so we might need many of these for the agent to understand how to generalize. This can slow down a lot the learning.

As we saw earlier, learning algorithms and agents’ models are heavily linked together. Indeed, the process of learning consists in modifying the model’s parameters. The algorithm can even modify the structure of the model. The choice of the learning algorithm or its design cannot be done without choosing or knowing the behavioral model. This makes this choice even more complicated as we must consider the pros and the cons for the couple and not for each one separately.

CONCLUSION

In this paper we outlined the problem of creating believable agents in video games. This problem has three parameters: the players, the video game and the agents.

We can add on top the learning mechanisms to make agents capable of evolution. All those factors have an important impact on the final result so their influence should be carefully studied.

First, believability is a very complex notion because of its subjectivity. We defined it as the “illusion that the agent is controlled by a player”. Although this definition is quite different from the one used in arts, it makes sense in video games. Indeed, several people can interact in the same game and replacing some of them by computer programs can be very interesting. What makes believability complex is that although players have difficulties in understanding the others’ behaviors and often interpret them wrongly, some small flaws can destroy the whole illusion.

The next important factor is the video game itself. A believable agent in an game where it is only possible to chat will be very different from a believable agent in an game where it is only possible to move. Complex games which require both reactive and planning abilities as well as featuring a wide range of possible interactions between players are the most challenging testbeds for researchers.

The agent itself and more precisely the model defining its behaviors is the most obvious factor in the problem. Models can express more or less believable behaviors depending on how they model human characteristics. So, although there are many models both in the industry and the research, they are not all suitable for believable agents.

An agent will not be able to make the illusion last very long if it is not capable of evolving. One possible way of evolution is learning. Imitation learning seems to fit perfectly with the goal of believability. Indeed, it consists in copying the demonstrator acts which is quite close to the notion of believability. It is also a very fast method for transmitting knowledge from one person to another.

The first step in the design of a believable agent is to define an evaluation method. This will help us in knowing which characteristics are important for our agent. This cannot be done without choosing a video game first because the notion of believability depends on the agent’s set of actions. It can be also very interesting to evaluate humans’ and simple agents’ believability first to better understand the problem.

The next step is to choose or design a behavioral model and an imitation learning algorithm. This should be done according to the believability factors discovered in the previous step. Considering the constrains of fast

以及使其更加显著的在线学习特性，这极大地减少了人工智能中通常可用的模型和算法。

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and online learning which make it noticeable, this reduces greatly the available models and algorithms usually used in artificial intelligence.

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